

MINH-HOANG LE

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EDUCATION

VNUHCM - University of Science

Oct 2022 – Nov 2026 (Expected)

Bachelor of Science - Advanced Program in Computer Science

GPA: 3.93/4

Honors: **Dean's List** of academic year 2024-2025 (**Top 5/118**)

Research Interests: Computer Vision, Deepfake Detection, Image Forensics, Vision-Language Models

PUBLICATIONS

Robust Deepfake Detection: Mitigating Spatial Attention Drift via Calibrated Complementary Ensembles, CVPRW 2026 (Co-first author, Accepted, Paper) **Jun 2026**

- Engineered a multi-stream, foundation-driven framework (integrating DINOv2 and CLIP) trained via an extreme-degradation pipeline to successfully ignore intentional noise (blur, compression) and anchor on core facial geometry.
- Placed **Rank 4/341** in the Robust Deepfake Detection Challenge @ CVPR 2026, employing a calibrated voting mechanism to mitigate spatial attention drift and deliver highly stable zero-shot generalization.

EDGER: EDge-Guided with HEatmap Refinement for Generalizable Image Forgery Localization, SoICT 2025 (Co-first author, Accepted, arXiv) **Dec 2025**

- Developed a resolution-agnostic, dual-branch framework that pinpoints AI-generated manipulations in any image size by fusing frequency-based edge boundaries and patch-level visual features.
- Demonstrated strong cross-domain generalization on multi-megapixel imagery in the MediaEval 2025 SynthIM challenge, ensuring high reliability for real-world forgery localization.

Patch-Based Localized Artifact Enhancement for Deepfake Detection, ICMV 2025 (Co-first author, Paper) **Oct 2025**

- Engineered a dual-branch deepfake detection model that examines both the overall face and strategically localized facial patches to spot subtle, hidden forgeries.
- Achieved a top-tier 91.4% average AUC, proving the system's robust cross-dataset generalization across five challenging benchmarks.

A Hybrid Model for Generalizable Deepfake Detection via Blending, Semantic, and General Artifacts, SCID Workshop, ACM AsiaCCS (Co-first author, Paper) **Aug 2025**

- Designed a comprehensive hybrid deepfake detector that identifies fake videos by simultaneously capturing unnatural blending, weird facial features, and general deepfake manipulation traces.
- Delivered state-of-the-art cross-dataset performance with a 90.8% video-level AUC across multiple benchmarks in recognizing unseen fake videos, making the system highly adaptable to new deepfake creation methods.

ENRIC: EveNt-AwaRe Captioning with Image Retrieval via UnCertainty-Guided Re-ranking and Semantic Ensemble Reasoning, ACM MM (Co-first author, Paper) **Jun 2025**

- Engineered an end-to-end pipeline integrating semantic retrieval via vision-language embeddings, uncertainty-guided re-ranking with confidence heuristics, and event-aware caption generation using Chain-of-Thought prompting to extract accurate news contexts for isolated images.
- Placed **Rank 1/84** in the EVENTA Grand Challenge @ ACM Multimedia 2025 with a top combined score of 0.5501, outperforming all competitors in generating accurate, context-aware narratives.

RESEARCH EXPERIENCE

SELAB - HCMUS

Nov 2024 – Present

Member

- Conducted research in Computer Vision, specifically focusing on Deepfake Detection, Image Forgery Localization, and cross-domain generalization under extreme zero-shot settings.
- Developed and evaluated complex deep learning architectures in PyTorch, resulting in multiple co-first-author publications in top-tier conferences and workshops.

AWARDS & ACHIEVEMENTS

Rank 51/141, ICPC World Finals 2024	Sep 2024
Rank 61/124, ICPC World Finals 2023	Apr 2024
Bronze Medal (Rank 9/64), The 2024 ICPC Asia Pacific Championship	Mar 2024
Silver Medal (Rank 8/125), The 2023 ICPC Asia Hue City Regional Contest	Dec 2023
Gold Medal (Rank 8/132), The 2022 ICPC Asia Ho Chi Minh City Regional Contest	Dec 2022

- Represented Vietnam to compete in the world's most prestigious algorithmic contest for college students, ranking among the **top 1% of global teams**.
- Demonstrated exceptional problem-solving skills and mastery of high-performance C/C++ by tackling complex, advanced algorithmic challenges under extreme time pressure.

Second Prize, Vietnamese Olympiad in Informatics (VOI) **2021, 2022**

- An annual programming contest held by the Vietnam Ministry of Education and Training for Vietnamese high-school students. This is the preliminary contest for choosing representatives for the International Olympiad in Informatics (IOI).

Second Prize, The Vietnam Student Artificial Intelligence Olympiad 2025 - Final Round **Dec 2025**

Third Prize, The Vietnam Student Artificial Intelligence Olympiad 2025 - Southern Regional **Nov 2025**

EXTRACURRICULAR ACTIVITIES

VNOI - Vietnam Olympiad in Informatics Community **Sep 2021 – May 2025**

Volunteer, Article Writing Team

- VNOI Wiki project publishes high-quality, free algorithmic articles, which are being read widely by students in Vietnam, to assist competitive programmers in learning algorithms and data structures. These articles attract an average of 24000 reads a month.
- Published 9 algorithmic articles: "Binary Indexed Tree", "LCA", "basic geometry 1", "basic geometry 2", "Convex Hull", "RMQ", "MITM", "Divide and Conquer DP", and "Convex Hull Trick".

TECHNICAL SKILLS

Languages: C/C++, Python, Java, Kotlin, Javascript, SQL

AI & Deep Learning: PyTorch, transformers, timm, OpenCV

Web & Tools: Git, Linux, ReactJS, Django, Selenium